We repeat the process one more time **(E)** for P6, and since its reachability distance (from P5) is smaller than ε, we know that it's reachable from P5 and we can add both to the same cluster **(F)**. If there were more points to process, we would have continued in the same way.

One final remark, before moving to the implementation, shown in listing 12.9: notice that, if we run OPTICS twice starting with the same point, the dataset will be processed in the same order, and the results will be identical (net of possible ties on reachability distance). Therefore, OPTICS can be considered as completely deterministic.

One note on the method `opticsCluster`: in the original paper, the authors presented a slightly different method using points' core distance to decide if a new cluster should be started. The version presented here stems from the consideration that, if the reachability of a point q is undefined or above the threshold ε, while point p, the successor to q in the processing order, has a reachability distance that's below ε, than q must be a core point: p's reachability distance from the set of points processed before q, in fact, must also be undefined or greater than q's, otherwise p would have been processed before q; consequently, p's reachability distance in the plot is the reachability distance between p and q, and according to section 12.4.1 that's only defined if q is a core point.

12.4.4 Hierarchical clustering

Now that we have learned how to produce a flat clustering from OPTICS results, we are in good shape to proficiently use